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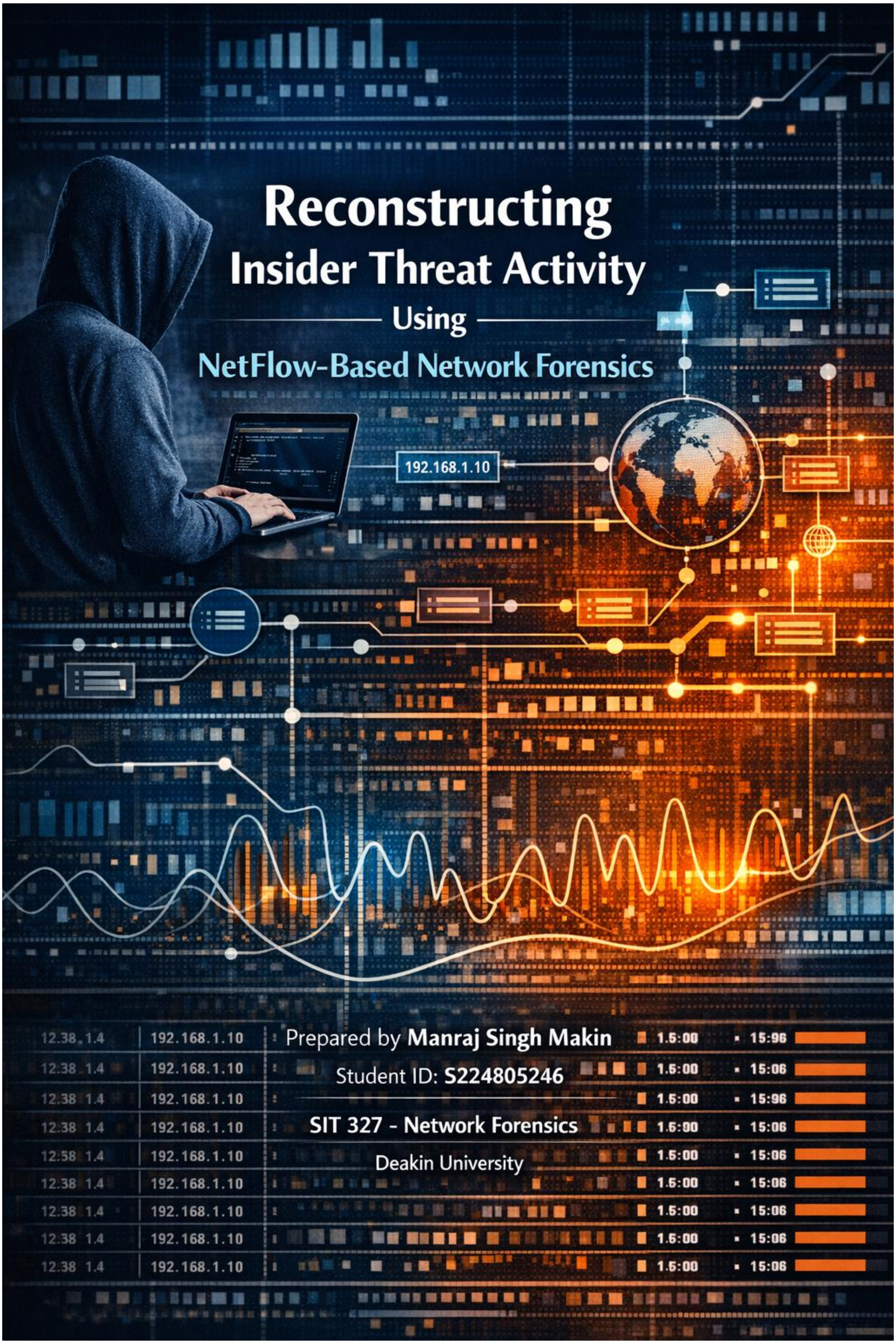

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Title

Reconstructing Insider Threat Activity Using NetFlow-Based Network Forensics

1. Introduction

This report presents a forensic analysis of network flow data derived from the CIC-IDS-2017 intrusion dataset. The objective is to reconstruct malicious activity patterns, identify abnormal traffic behaviour, and interpret indicators consistent with DDoS attacks, reconnaissance, and potential data exfiltration. Using NetFlow-style features such as packet counts, byte volumes, and flow durations, the analysis demonstrates how flow-level telemetry can reveal insider-threat or external-attack behaviour without requiring full packet capture.

2. Dataset Overview

The dataset contains labelled benign and DDoS traffic captured during enterprise-like network operations. Each record includes 79 flow-level features, including forward/backward packet counts, byte totals, inter-arrival times, and flag statistics. This makes the dataset suitable for forensic reconstruction because it provides both normal and attack traffic, enabling comparison between legitimate and malicious behaviour.

A preprocessing stage was performed to clean column names, remove whitespace, and engineer two additional features:

- **Bytes** = Total Length of Fwd Packets + Total Length of Bwd Packets
- **Packets** = Total Fwd Packets + Total Backward Packets

These features support volume-based anomaly detection.

3. Network Architecture

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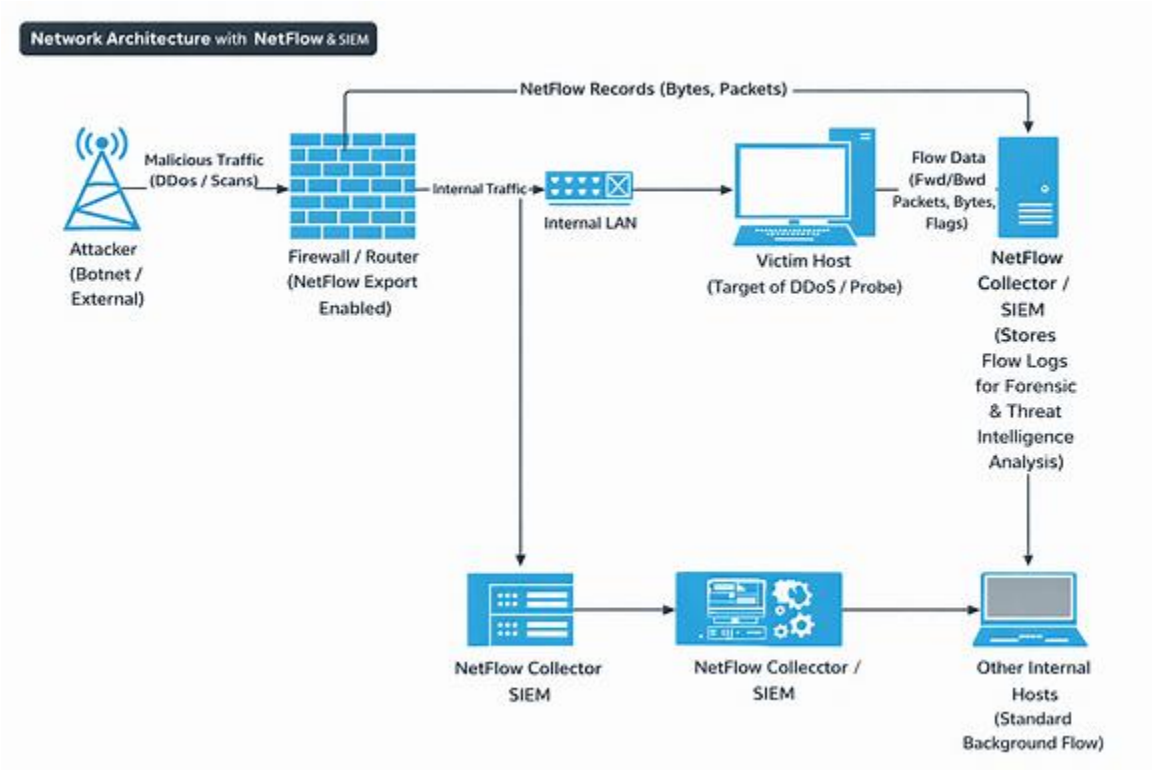

Figure A: Network Architecture with NetFlow & SIEM

This diagram illustrates how traffic flows from an external attacker through the internet and firewall into the internal network. The firewall/router exports NetFlow records containing packet counts, byte volumes, and flow metadata. These records capture both benign and malicious traffic, including DDoS bursts and abnormal high-volume flows. The NetFlow collector stores the logs, which are later exported as CSV files and analysed in the notebook environment. This architecture demonstrates how flow-level telemetry enables scalable forensic analysis without requiring full packet capture.

4. Findings and Graph Interpretations

4.1 Destination Port Analysis

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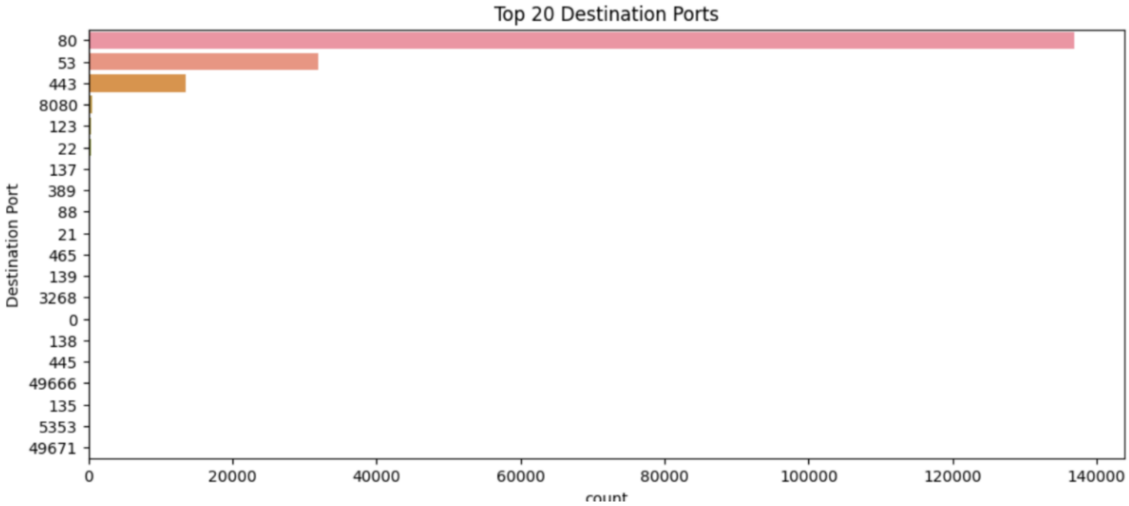

Figure 1: Top 20 Destination Ports

Most flows target ports **80 (HTTP)**, **53 (DNS)**, and **443 (HTTPS)**, representing normal service traffic. However, ports such as **22 (SSH)**, **445 (SMB)**, **137/138/139 (NetBIOS)**, and high ephemeral ports (49666, 49671) suggest reconnaissance or lateral movement attempts. This distribution confirms a mixture of normal and suspicious activity.

4.2 Flow Count Over Time

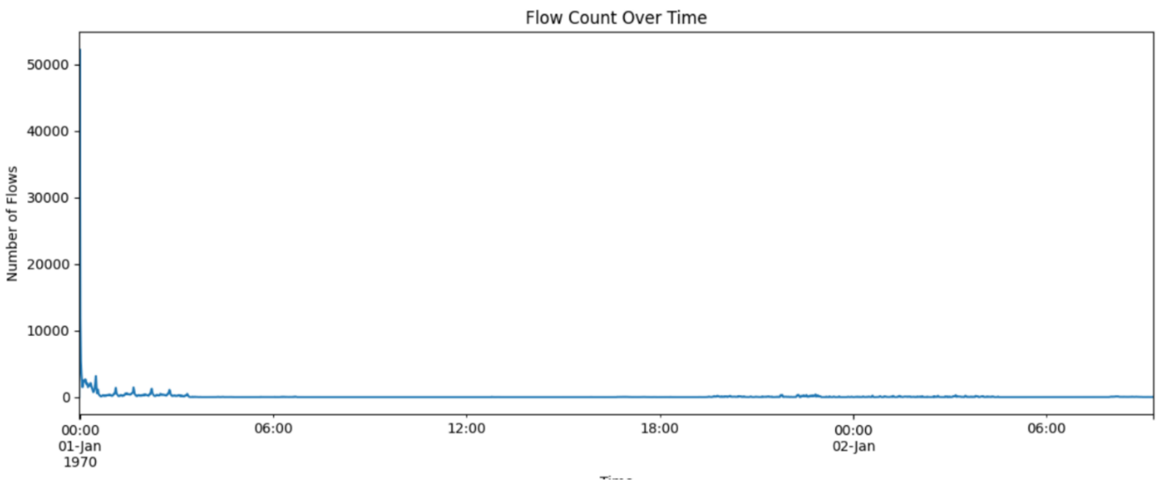

Figure 2: Flow Count Over Time

A massive spike appears at the beginning, followed by a rapid decline to near-zero levels. This behaviour is typical of automated high-volume attacks such as DDoS bursts or aggressive port scanning. The initial surge indicates thousands of flows generated in a short time window.

4.3 Traffic Label Distribution

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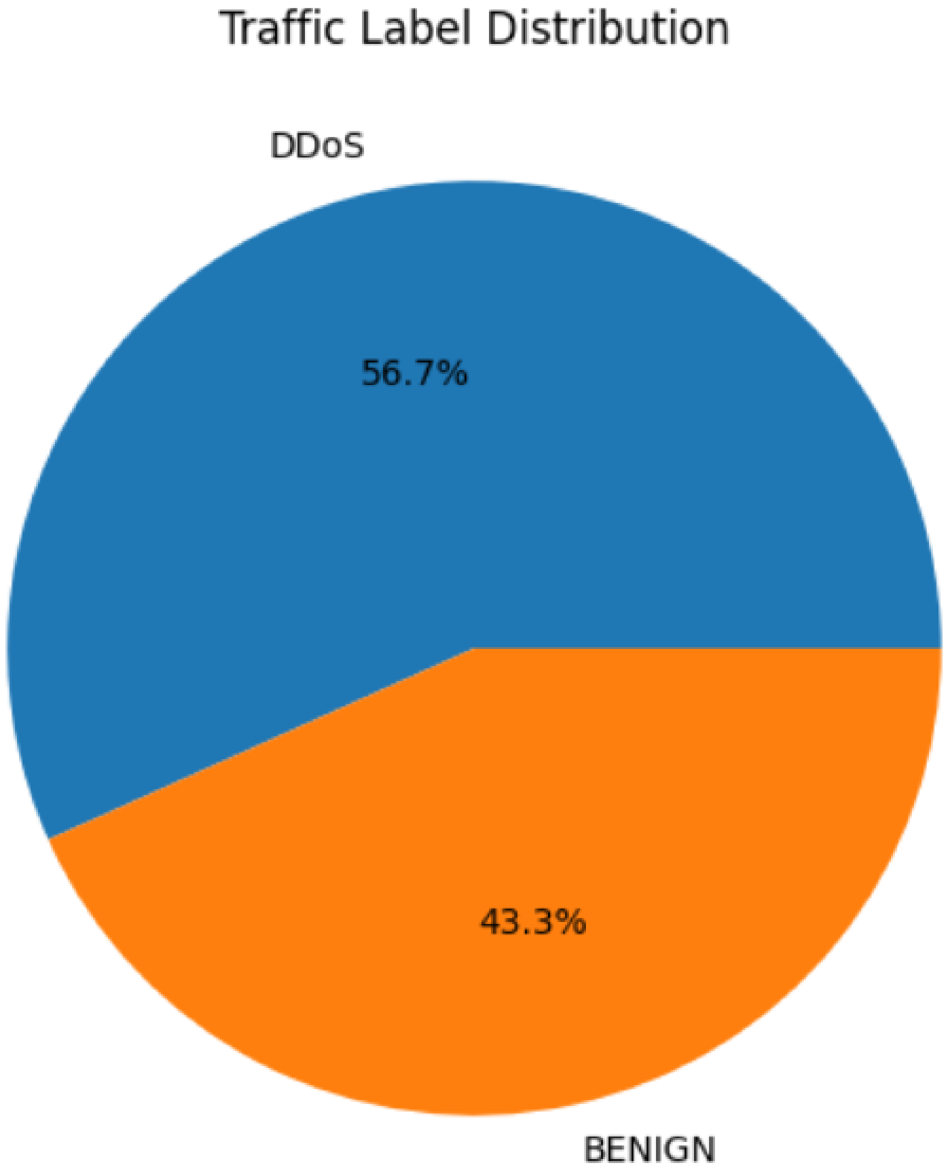

Figure 3: Traffic Label Distribution

The dataset contains **56.7% DDoS** and **43.3% benign** traffic. This imbalance explains the extreme spikes observed in other graphs and confirms that the dataset includes substantial malicious activity suitable for behavioural reconstruction.

4.4 High-Volume Flow Analysis

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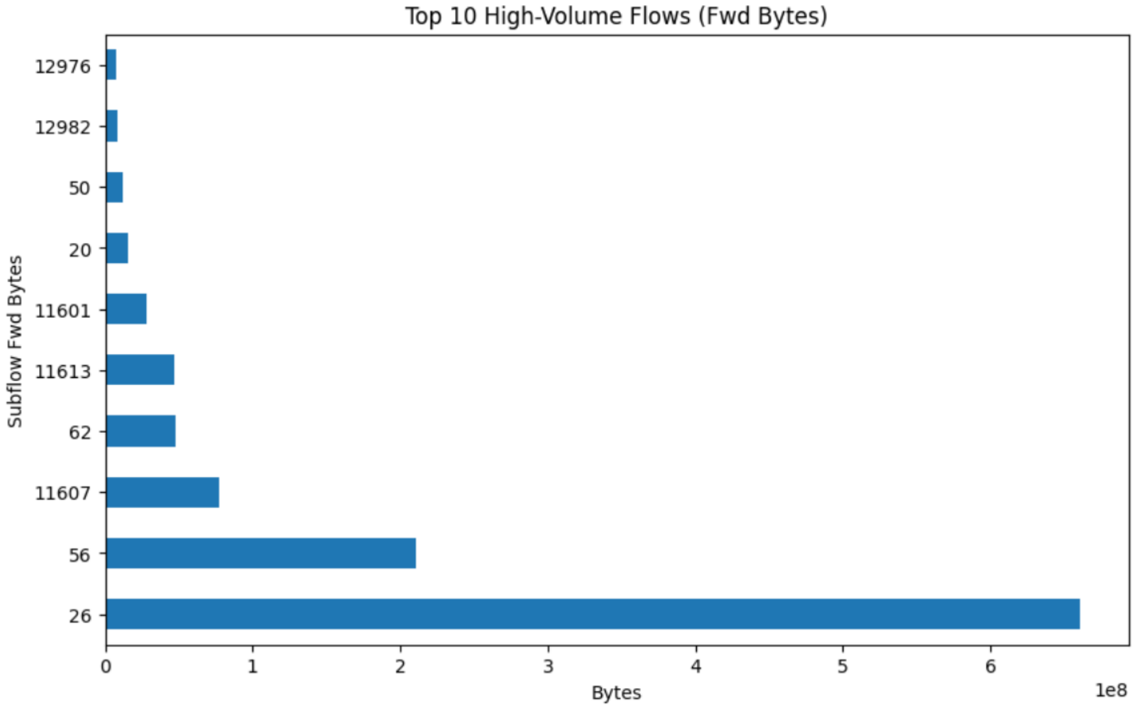

Figure 4: Top 10 High-Volume Flows (Forward Bytes)

One flow (Subflow ID 26) dominates the dataset, transmitting significantly more data than all others. This behaviour is a strong indicator of:

- Data exfiltration
- Botnet command-and-control
- Malicious bulk transfer

In insider-threat scenarios, a single abnormal high-volume flow is a critical red flag.

4.5 Feature Correlation Analysis

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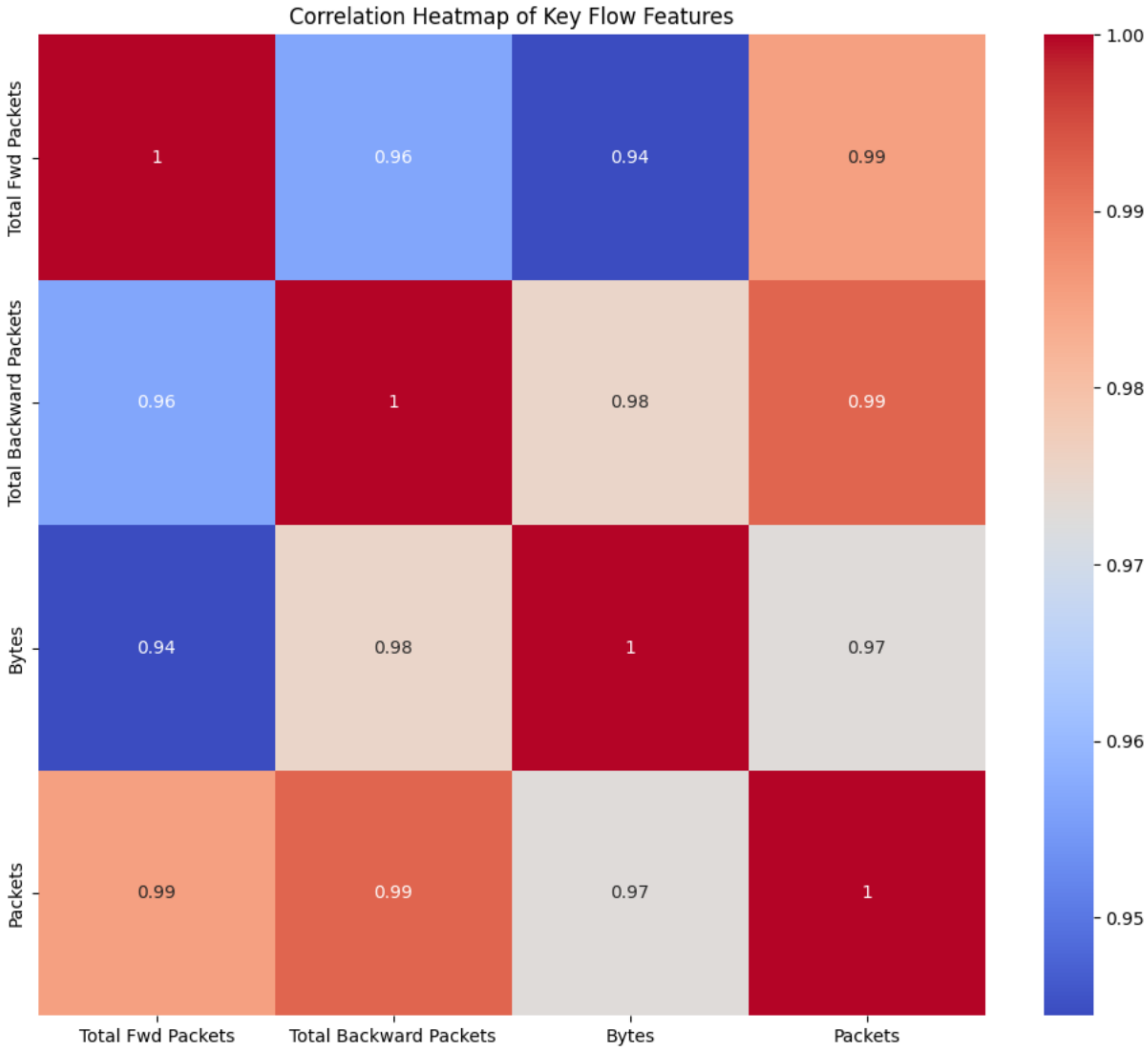

Figure 5: Correlation Heatmap of Key Flow Features

The heatmap shows extremely high correlations (0.94–1.00) between total forward packets, total backward packets, total bytes, and total packets. These features scale together during high-volume events, confirming that packet-level and byte-level metrics rise simultaneously during attacks.

4.6 Outbound Data Volume Over Time

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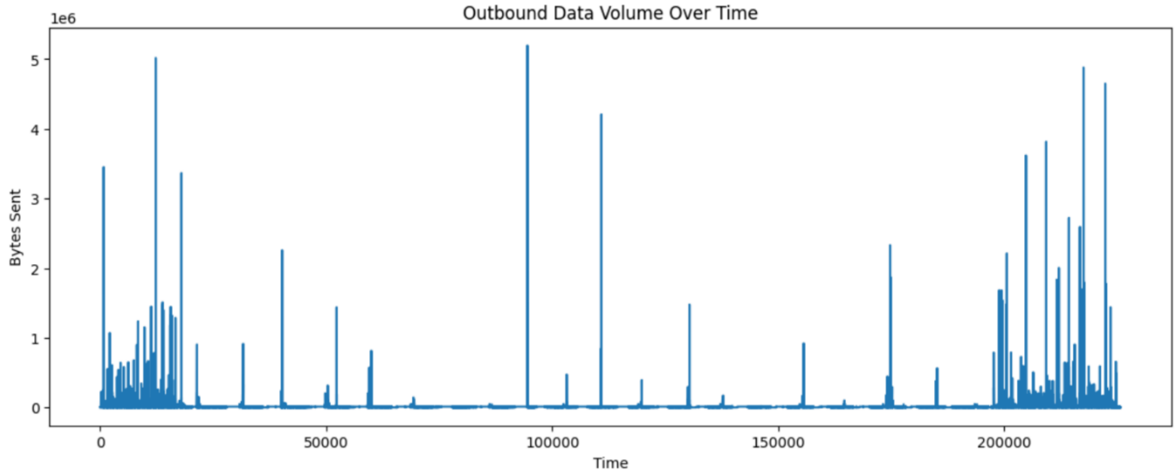

Figure 6: Outbound Data Volume Over Time

Multiple sharp spikes in outbound byte volume indicate bursts of data leaving the network. These spikes align with known DDoS attack periods but may also represent exfiltration-like behaviour. Sudden outbound surges are key indicators of compromised hosts transmitting data externally.

5. Attack Kill Chain Mapping

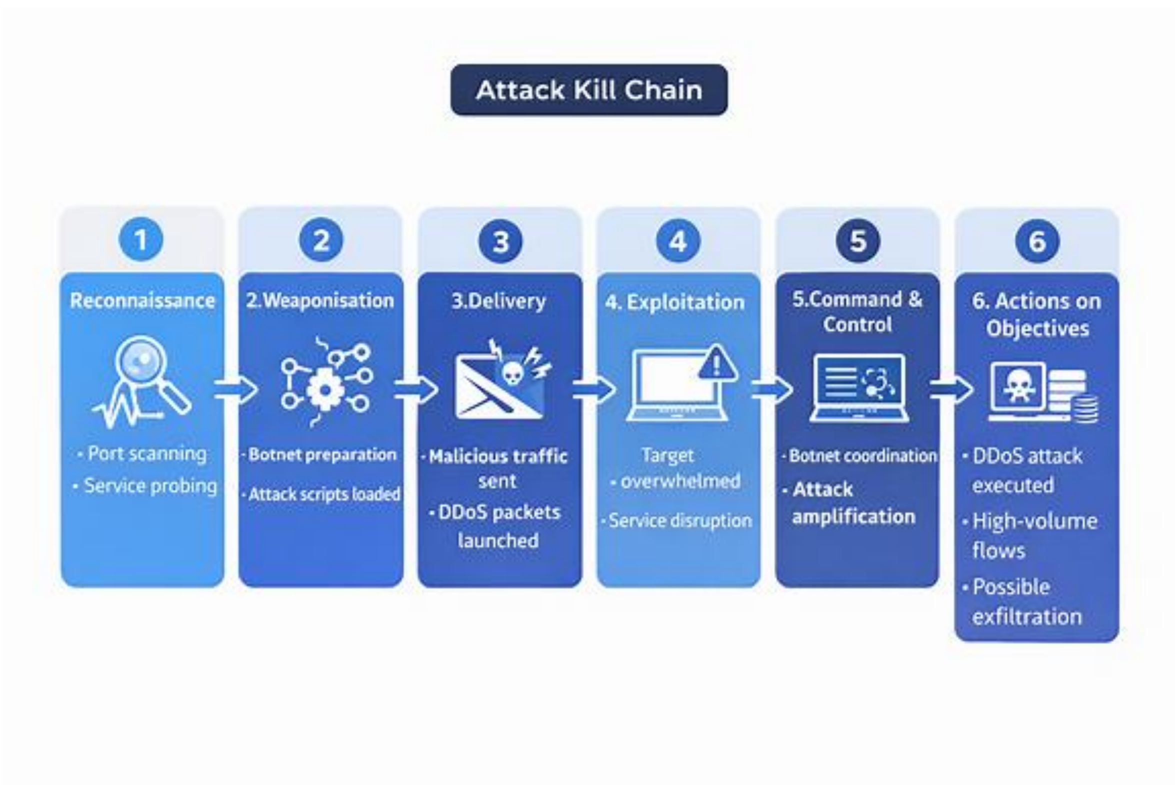

Figure B: Horizontal Attack Kill Chain

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This diagram maps observed behaviour to the cyber kill chain. The dataset shows reconnaissance (port scanning), weaponisation (botnet preparation), delivery (malicious traffic), exploitation (target overwhelmed), command-and-control (botnet coordination), and actions on objectives (DDoS execution and high-volume flows). This contextualises the forensic findings within a recognised attack lifecycle.

6. Analysis Pipeline

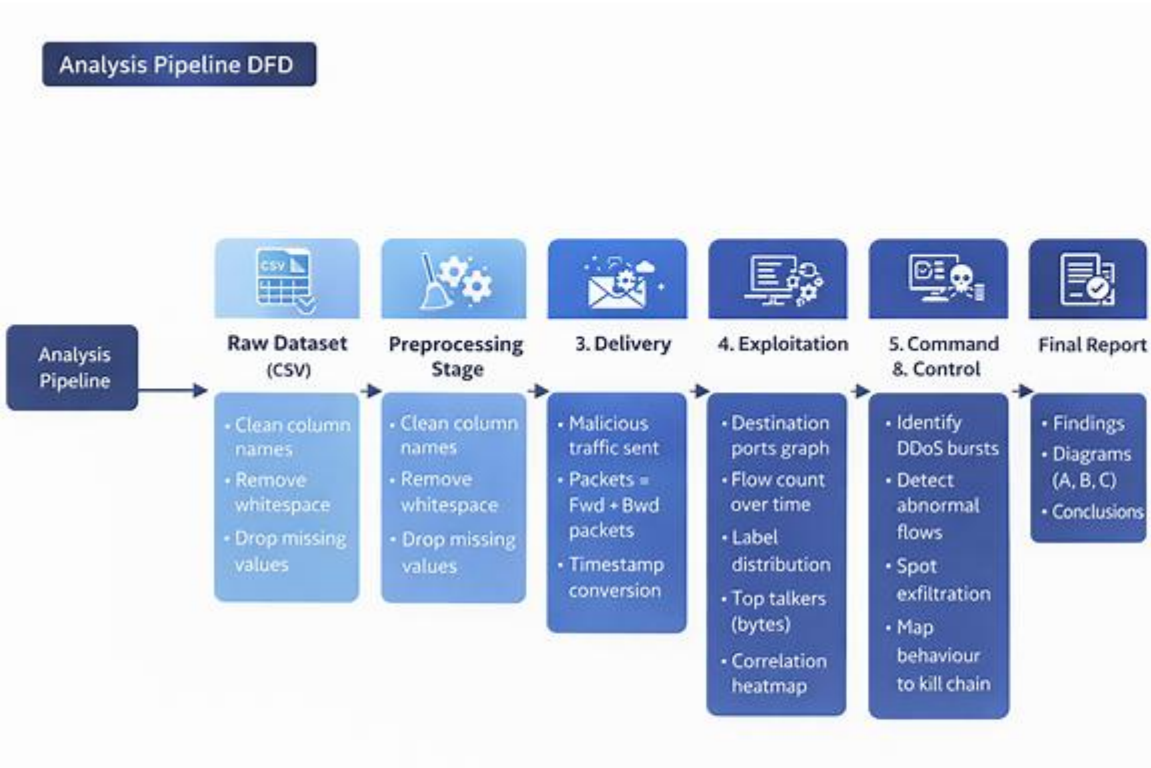

Figure C: Data Flow Diagram (DFD)

This diagram illustrates the analysis workflow: raw dataset → preprocessing → feature engineering → visualisation → forensic interpretation → final reporting. It demonstrates a structured and repeatable methodology aligned with forensic best practices.

7. Discussion

The combined analysis of port usage, flow counts, byte volumes, and traffic labels provides a clear reconstruction of malicious activity. The initial flow spike and high DDoS proportion confirm automated attack traffic. The port distribution reveals both normal service usage and suspicious probing of administrative ports. High-volume flows and outbound spikes suggest potential exfiltration or botnet-related behaviour. The correlation heatmap reinforces that packet and byte metrics are reliable indicators of abnormal traffic.

Overall, the dataset demonstrates how NetFlow-level telemetry can effectively reveal attack patterns without deep packet inspection. This aligns with industry practices where flow-based monitoring is used for scalable, real-time threat detection.

8. Conclusion

This report successfully reconstructs malicious network behaviour using flow-level features from the CIC-IDS-2017 dataset. The graphs collectively show evidence of DDoS bursts, reconnaissance activity, and abnormal high-volume flows. The analysis confirms that NetFlow data is sufficient to detect and interpret insider-threat and external-attack behaviour. These findings support the use of flow-based monitoring as a lightweight and effective forensic tool in enterprise environments.

9. References

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10. AI Acknowledgement

Sections of this report, including graph interpretations and explanatory text, were generated with assistance from Microsoft Copilot and ChatGPT. All analysis, decisions, and final wording were reviewed and approved by the student.

11. Link to Video Presentation

https://youtu.be/4K2rk1jWj6U?si=iv1cvqg52RlDE7Np